

INFRARED SENSOR (LM393) MODULE

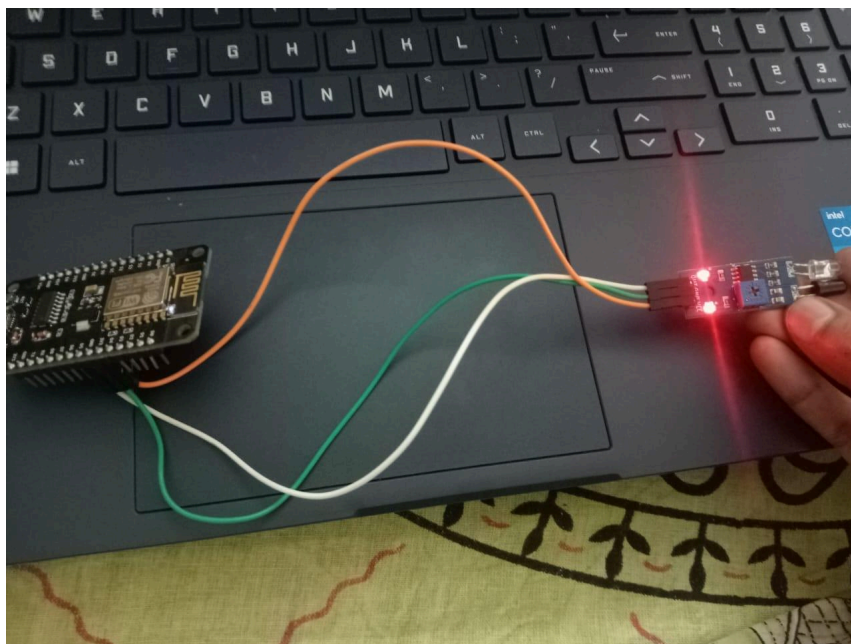
Introduction:

This document demonstrates how to connect an IR sensor to an ESP8266 microcontroller, use the Arduino IDE to upload the code, and visualize the output using the Serial Plotter, Serial Monitor, and Mixed Signal Oscilloscope (MSO). The purpose is to generate a straight line when no object is detected and a waveform when an object is detected.

Components Required:

1. ESP8266 Microcontroller (e.g., NodeMCU)
2. Infrared Sensor Module LM393 Module
3. MSO (Mixed Signal Oscilloscope)
4. Jumper Wires
5. USB Cable
6. Arduino IDE

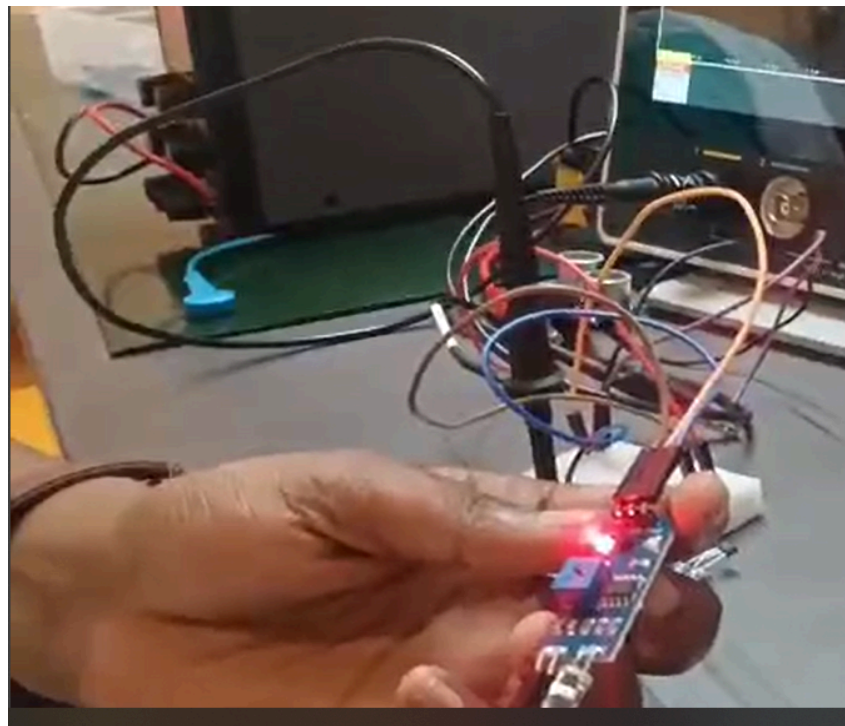
Circuit Diagram:



Connections:

Infrared sensor module LM393 Pin	ESP8266 Pin
VCC	3.3V
GND	GND
OUT	D5(GPIO 14)

MSO Circuit Diagram:



MSO Connections:

Mixed Signal Oscilloscope	Infrared sensor module LM393 Pin
MSO Pin	Connection Point
GND	GND
Signal IN	OUT

Code Explanation:

1. Initialize Serial communication at 115200 baud rate.
2. Set up the IR sensor pin as an input.
3. Continuously read the sensor value and print it to the Serial Monitor and Serial Plotter.
4. Generate a straight line when no object is detected and a waveform when an object is detected.

Arduino Code:

```
Unset
// IR Sensor with ESP8266 - Serial Monitor and Plotter

const int irPin = 14;  // GPIO 14 (D5)
int sensorValue = 0;

void setup() {
  Serial.begin(115200);  // Initialize Serial Monitor
  pinMode(irPin, INPUT);  // Set IR sensor pin as input
}

void loop() {
  sensorValue = digitalRead(irPin);  // Read sensor value
```

Serial Monitor Output:

Serial Plotter Output:

COM5

Interpolate ☐ STOP

value 1

The graph displays a constant signal value of 0 over the time interval from 3062 ms to 3111 ms. The y-axis is labeled from -1.5 to 1.5 in increments of 0.5. The x-axis is labeled with values 3062, 3074, 3086, 3098, and 3111.

Time (ms)	Value
3062	0
3074	0
3086	0
3098	0
3111	0

Type Message SEND New Line 115200 baud

Plot when object is detected:



MSO Output Observation:

MSO output when no object is detected:



MSO output when object is detected:



Troubleshooting:

1. If no output appears, check the wiring and ensure the correct COM port is selected.
2. If the graph is not appearing, adjust the threshold value according to your environment.
3. Make sure to use the correct GPIO pin for the IR sensor.

Video Link: [link](#)

INTEGRATING INFRARED SENSOR OUTPUT WITH GOOGLE SHEETS

Step 1: Setting Up the IR Sensor with ESP8266

IR Sensor Connections:

- VCC to 3.3V (ESP8266)
- GND to GND (ESP8266)
- Output Pin (IR) to D2 (ESP8266)

Code:

1. Open Arduino IDE.
2. Install required libraries:
 - ESP8266WiFi.h
 - ESP8266HTTPClient.h
3. Write the following code:

Unset

```
#include <ESP8266WiFi.h>
#include <ESP8266HTTPClient.h>

const char* ssid = "yourSSID";
const char* password = "yourPassword";
const int irPin = D2;

// Google Sheets Script URL
String url = "https://script.google.com/macros/s/YourScriptID/exec";

void setup() {
  Serial.begin(115200);
  WiFi.begin(ssid, password);
  pinMode(irPin, INPUT);
}
```

```
while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
}
Serial.println("WiFi Connected");
}

void loop() {
    int sensorValue = digitalRead(irPin);
    Serial.println(sensorValue);

    if (WiFi.status() == WL_CONNECTED) {
        HTTPClient http;
        String fullURL = url + "?detected=" + String(sensorValue);
        http.begin(fullURL);
        int httpCode = http.GET();
        Serial.print("HTTP Response code: ");
        Serial.println(httpCode);
        http.end();
    }
    delay(2000);
}
```

Step 2: Setting Up Google Sheets

Creating a Google Script:

1. Open Google Sheets.
2. Navigate to **Extensions > Apps Script**.
3. Copy and paste the following script:

```
Unset
function doGet(e) {
```



```
var sheet = SpreadsheetApp.openById("YourSpreadsheetID").getActiveSheet();
var detected = e.parameter.detected;
sheet.appendRow([new Date(), detected]);
return ContentService.createTextOutput("Data recorded");
}
```

4. Deploy the script as a web app with access to anyone.
5. Copy the script URL and replace it in the Arduino code.

Step 3: Testing and Debugging

1. Connect the ESP8266 to your computer.
2. Upload the code.
3. Open Serial Monitor to check Wi-Fi connectivity and sensor readings.
4. Observe data being logged in Google Sheets.

Step 4: Troubleshooting

Step 4: Troubleshooting

- **Wi-Fi Not Connecting:** Check SSID and password.
- **Data Not Logging:** Check the Google Script URL and permissions.
- **Serial Monitor Shows Errors:** Check connections and sensor status.
- **Library Not Found:** Make sure all required libraries are properly installed in Arduino IDE.
- **ESP8266 Not Responding:** Try resetting the module or using a different USB cable.

- **Google Sheets Not Updating:** Verify the script URL and check for script errors in Google Apps Script.
- **Sensor Giving Constant Values:** Inspect the sensor connections and ensure it is properly aligned.

OUTPUT:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
77	4/1/2025 16:02:59	DETECTION												
78	4/1/2025 16:03:05	1												
79	4/1/2025 16:03:11	1												
80	4/1/2025 16:03:17	1												
81	4/1/2025 16:03:23	1												
82	4/1/2025 16:03:29	1												
83	4/1/2025 16:03:34	1												
84	4/1/2025 16:03:41	1												
85	4/1/2025 16:03:47	1												
86	4/1/2025 16:03:55	1												
87	4/1/2025 16:04:01	1												
88	4/1/2025 16:04:06	1												
89	4/1/2025 16:04:12	1												
90	4/1/2025 16:04:19	1												
91	4/1/2025 16:04:24	1												
92	4/1/2025 16:04:30	1												
93	4/1/2025 16:04:36	1												
94	4/1/2025 16:04:42	1												
95	4/1/2025 16:04:48	1												
96	4/1/2025 16:04:54	1												
97	4/1/2025 16:05:00	1												
98	4/1/2025 16:05:06	1												
99	4/1/2025 16:05:12	1												
100	4/1/2025 16:05:18	1												
101	4/1/2025 16:05:25	1												
102	4/1/2025 16:05:31	1												
103	4/1/2025 16:05:37	1												
104	4/1/2025 16:05:43	1												
105	4/1/2025 16:05:49	0												
106	4/1/2025 16:05:54	0												
107	4/1/2025 16:06:00	0												
108	4/1/2025 16:06:05	0												
109	4/1/2025 16:06:11	0												
110	4/1/2025 16:06:17	0												
111														

Conclusion

This project successfully logs IR sensor data to Google Sheets using the ESP8266. The real-time data can be used for various applications, including motion detection and object counting.